L-bracket: ORIGINAL SOLID KEPT

```
vshell run summary: L-bracket
================================================================================

decision
--------
  outcome                         original SOLID part kept
  reason                          shelling does not improve the part: combined
                                  gain J = +0.0043 is not above ln(1 + 0.02)
  best design (k solid regions)   1
  final mesh                      final_design.stl

part
----
  STEP file                       /home/claude/vshell_repo/examples/step/Solid
                                  L-bracket.step
  volume (B-rep) mm3              4.198e+04
  volume (field) mm3              4.196e+04
  mass g                          43.66
  faces                           127
  analytic faces                  34
  support faces                   [9, 16, 24]
  load faces                      [[37]]
  force N                         [0, 0, -200]

segmentation
------------
  joints                          6
  struts                          9
  mounts                          4
  rank order (most shear-critical first) [6, 5, 14, 13, 1, 4, 11, 12, 2, 15, 3, 10, 8,
                                  9, ...]

baseline (solid)
----------------
  stiffness K0 N/mm               48.25
  load capacity P0 N              540.3
  governing mode                  strength
  lambda strength                 2.701
  lambda Euler                    5.169
  von Mises q MPa                 13.55
  mount von Mises q MPa           20.45
  FE DOF                          341463
  FE cells                        93372
  CG iterations                   37

best hollow candidate
---------------------
  solid regions                   [6, 16, 17, 18, 19]
  hollow regions                  [1, 2, 3, 4, 5, 7, 8, 9, 10, 11, 12, 13, 14,
                                  15]
  outer offset / strut wall s mm 1.805
  as-built wall mm                1.605
  stiffness K N/mm                51.54
  K/K0                            1.068
  load capacity P N               510.2
  P/P0                            0.9443
  J                               0.004342
  governing mode                  strength
  governing region                -
  lambda strength / Euler / local [2.551, 15.96, 19.06]
  mass (field) g                  43.64
  enclosed voids                  1
  filled small voids mm3          0
  mesh volume mm3                 4.194e+04
  mesh watertight                 no

search
------
  hollow designs analysed         10
  feasible in mass sweep          15
  objective                       {'w_stiffness': 0.5, 'w_capacity': 0.5,
                                  'min_gain': 0.02, 'floor_stiffness': 0.85,
                                  'floor_capacity': 0.85}

grid
----
  sdf_mm                          0.4
  shape                           [336, 476, 222]
  fe_mm                           0.8
  seg_mm                          1
```

n_exact	2885716
n_active	2360095

timings s

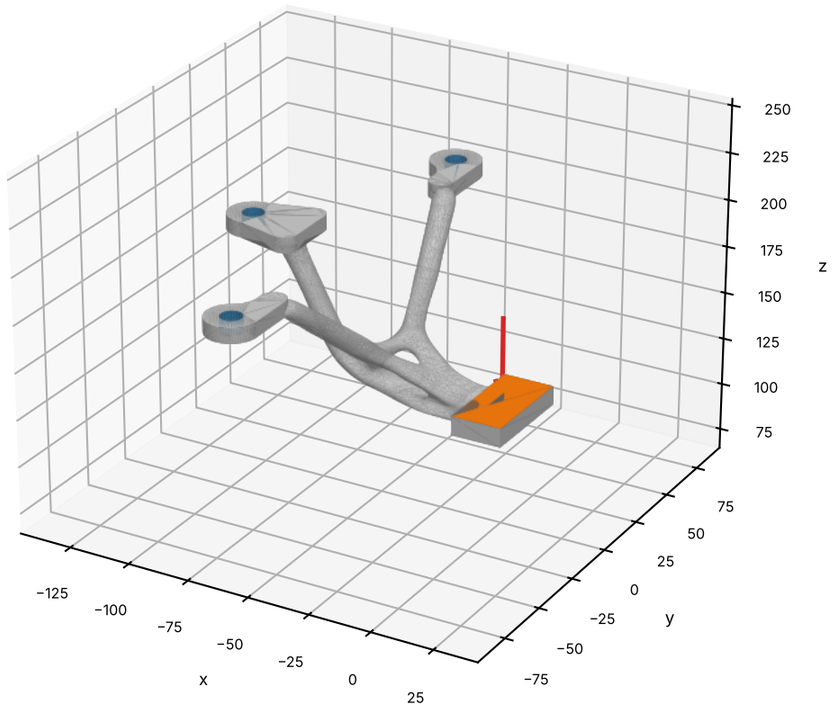
geometry	2.561
sdf	235.2
segmentation	11.79
fe	413.7
mass_sweep	82.36
outputs	5.154
total	831

provenance

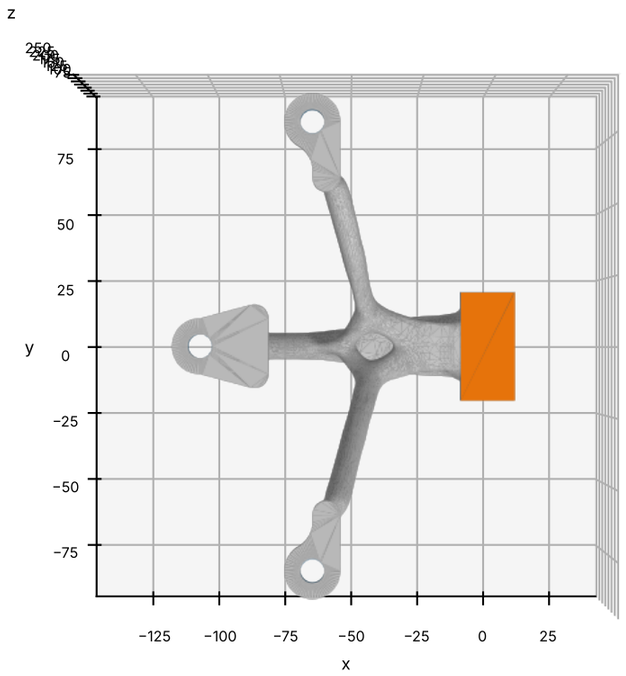
time (UTC)	2026-10-28T08:25:34+00:00
host	vm
platform	Linux-6.18.44-fc-v180-x86_64-with-glibc2.39
python	3.13.18
git	-
STEP sha256	b48aef84085ec4a9
packages	numpy 2.5.3, scipy 1.18.1, scikit-image 0.26.0, skan 0.13.1, libigl 2.6.3, pyamg 5.3.0, trimesh 5.2.0, matplotlib 3.11.2, build123d 0.13.0

L-bracket: supports (blue, all DOF fixed) and load (orange faces, red arrow, |F| = 200 N)

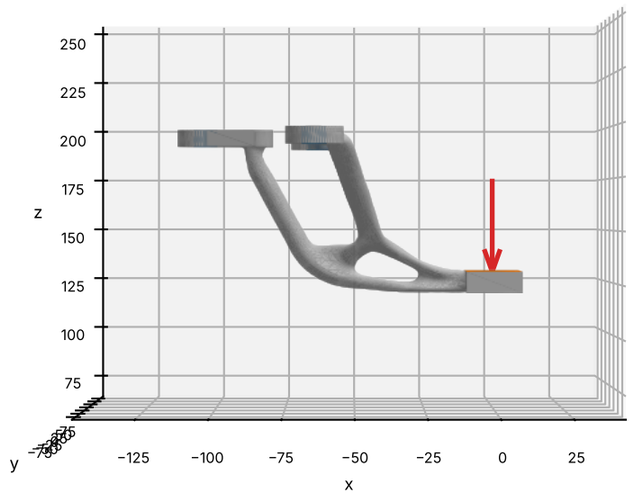
isometric



top (+z)

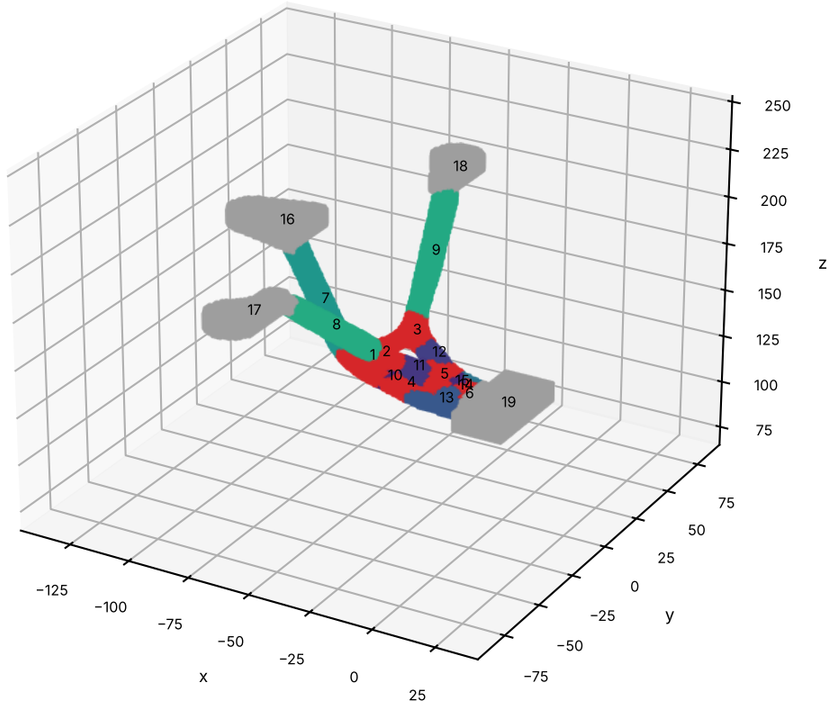


front (-y)

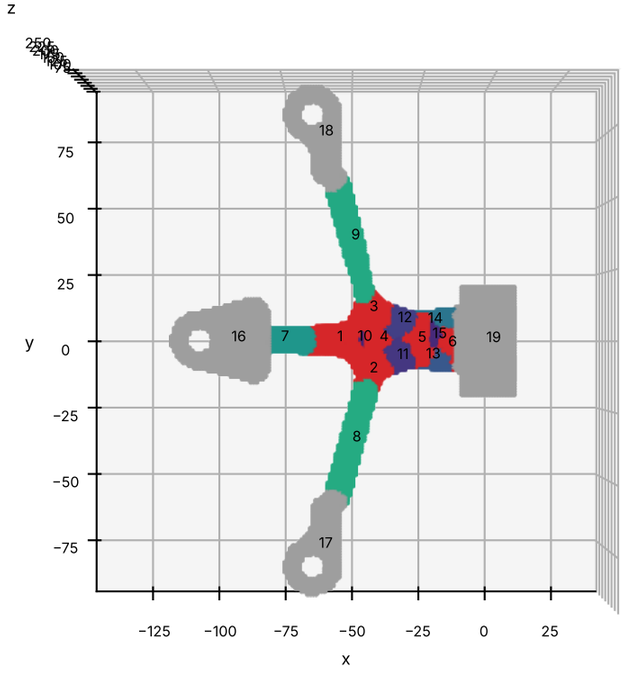


Segmentation: 6 joints (red), 9 struts (colour = slenderness lambda, 0 to 60+), 4 mounts (grey, never hollowed); numbers = region ids

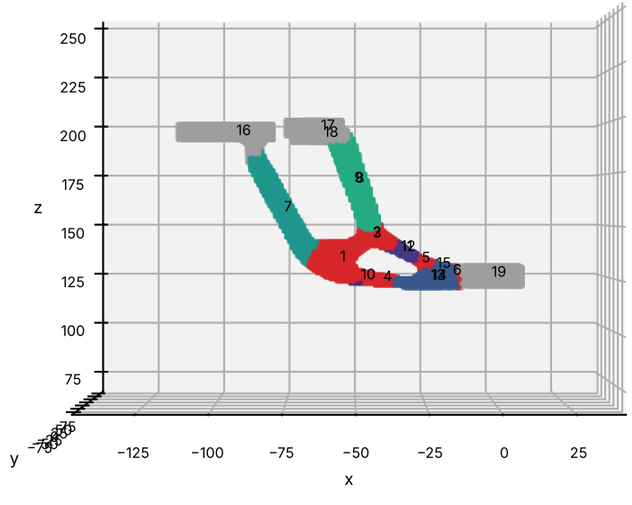
isometric



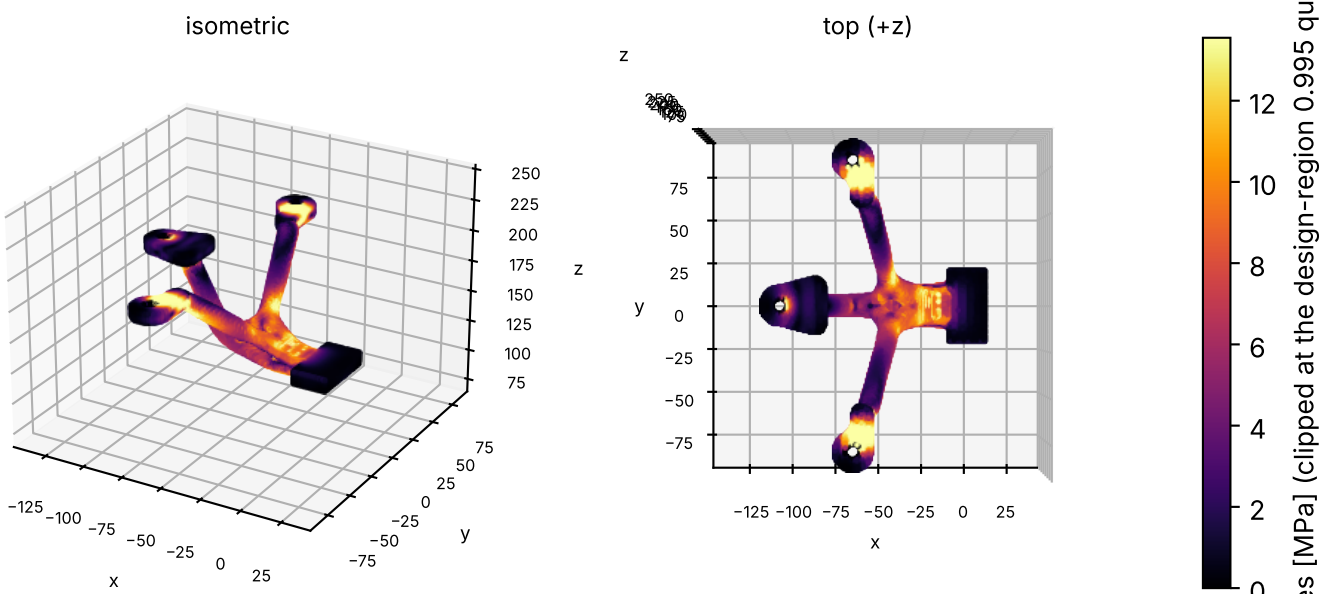
top (+z)

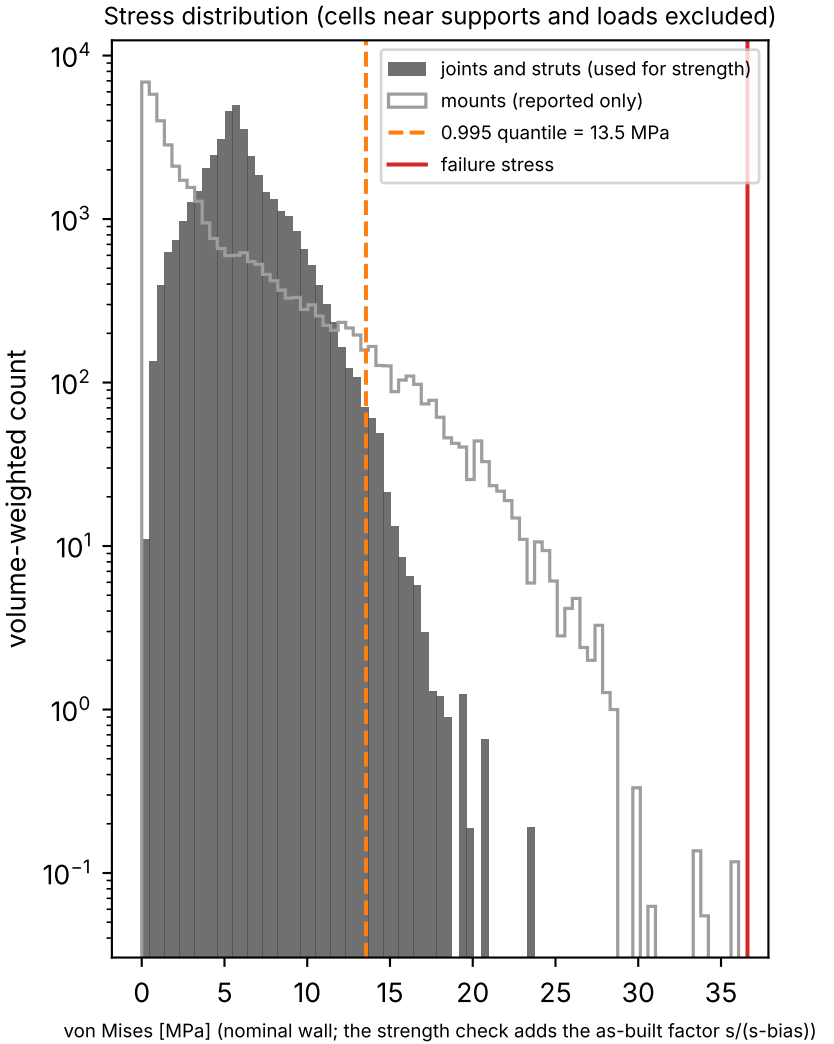


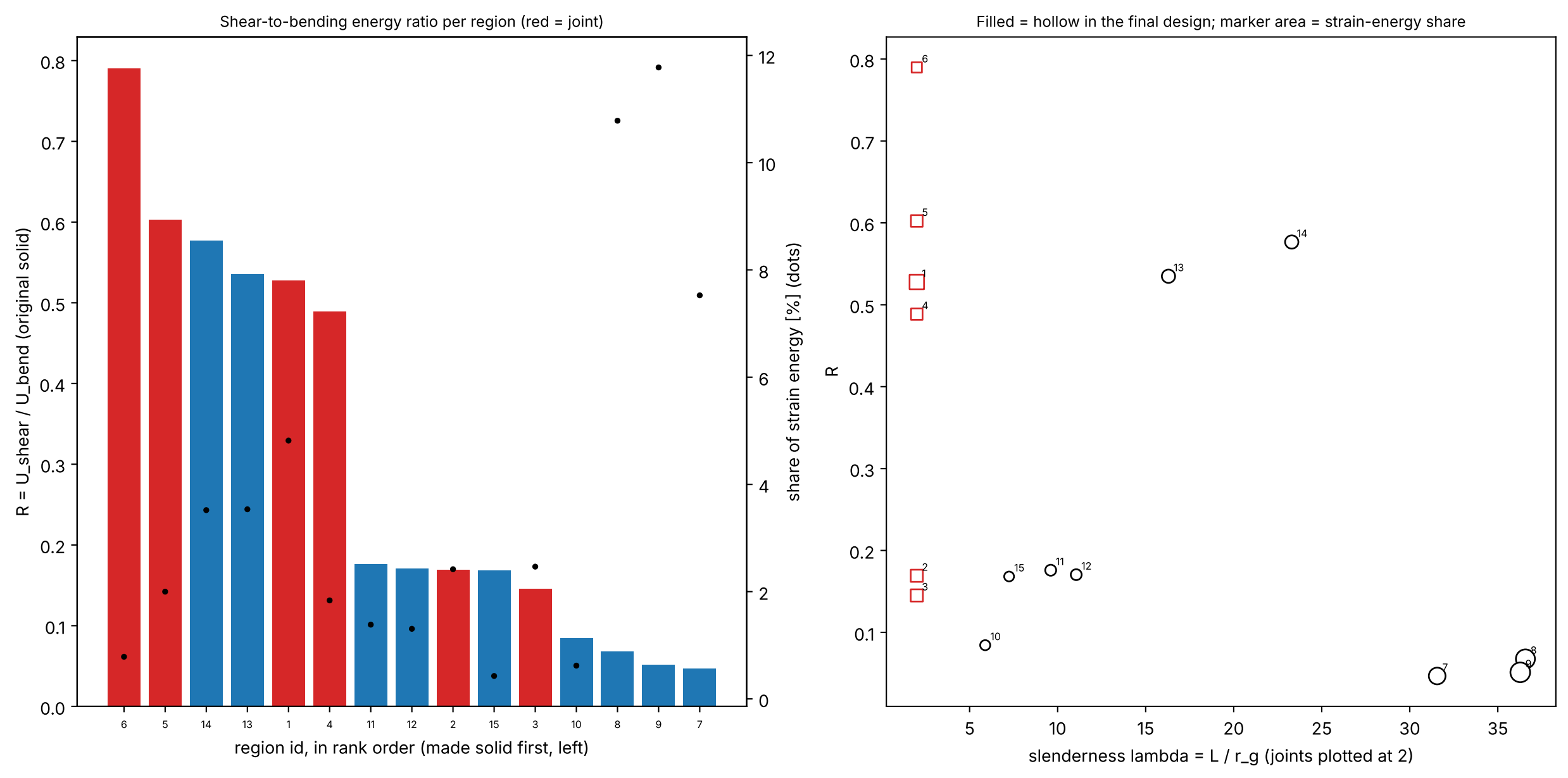
front (-y)

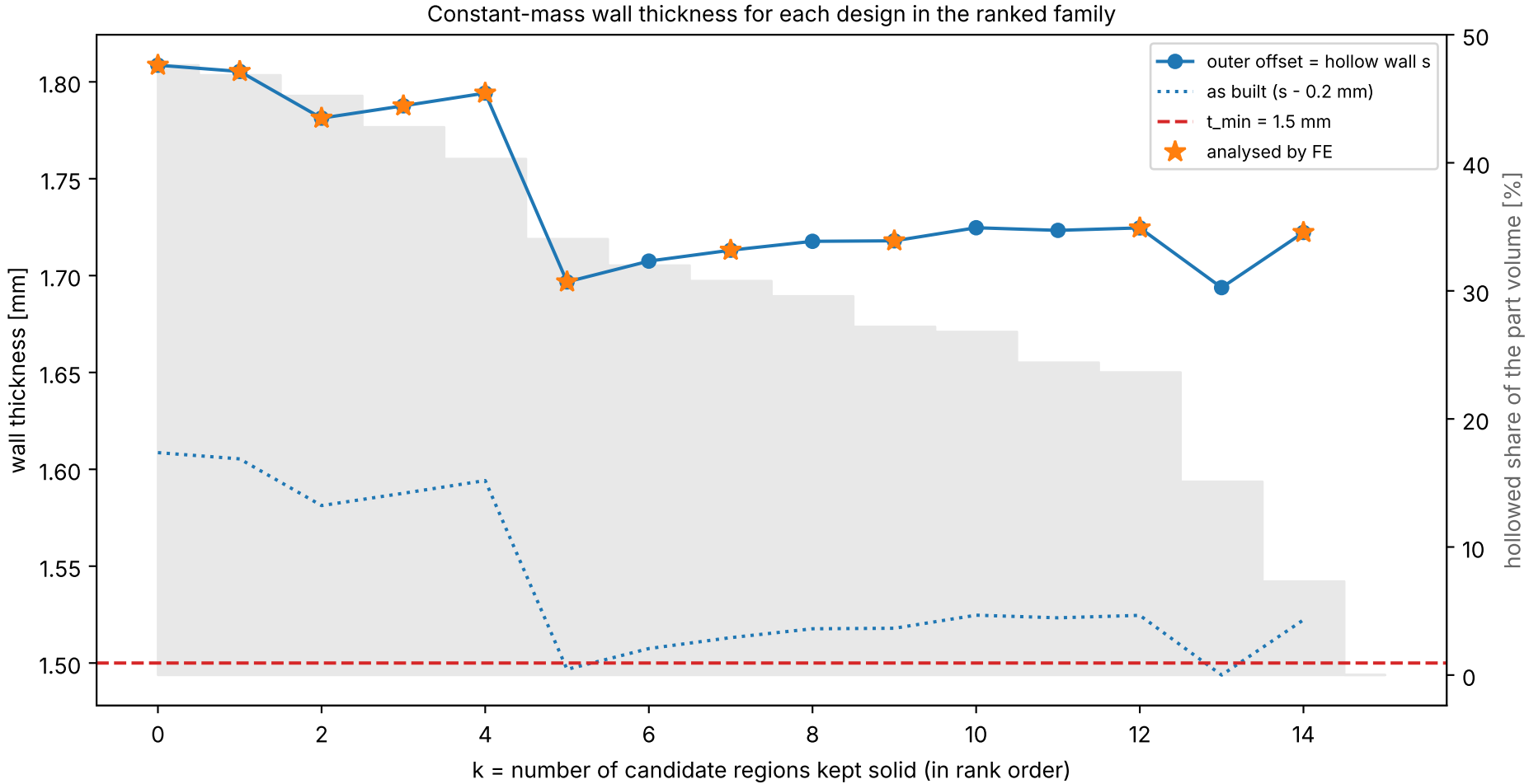


Original solid: von Mises stress at |F| = 200 N



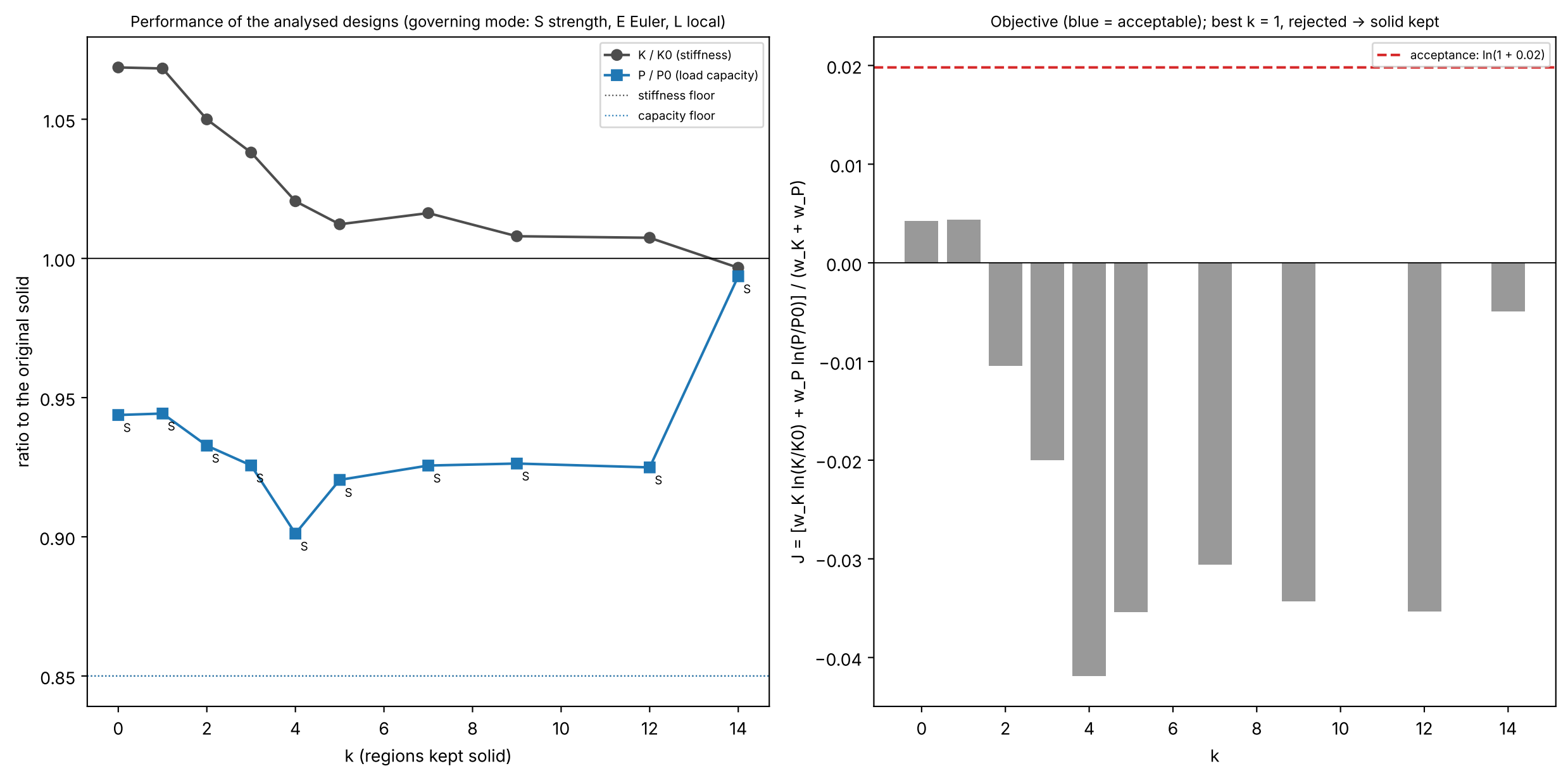


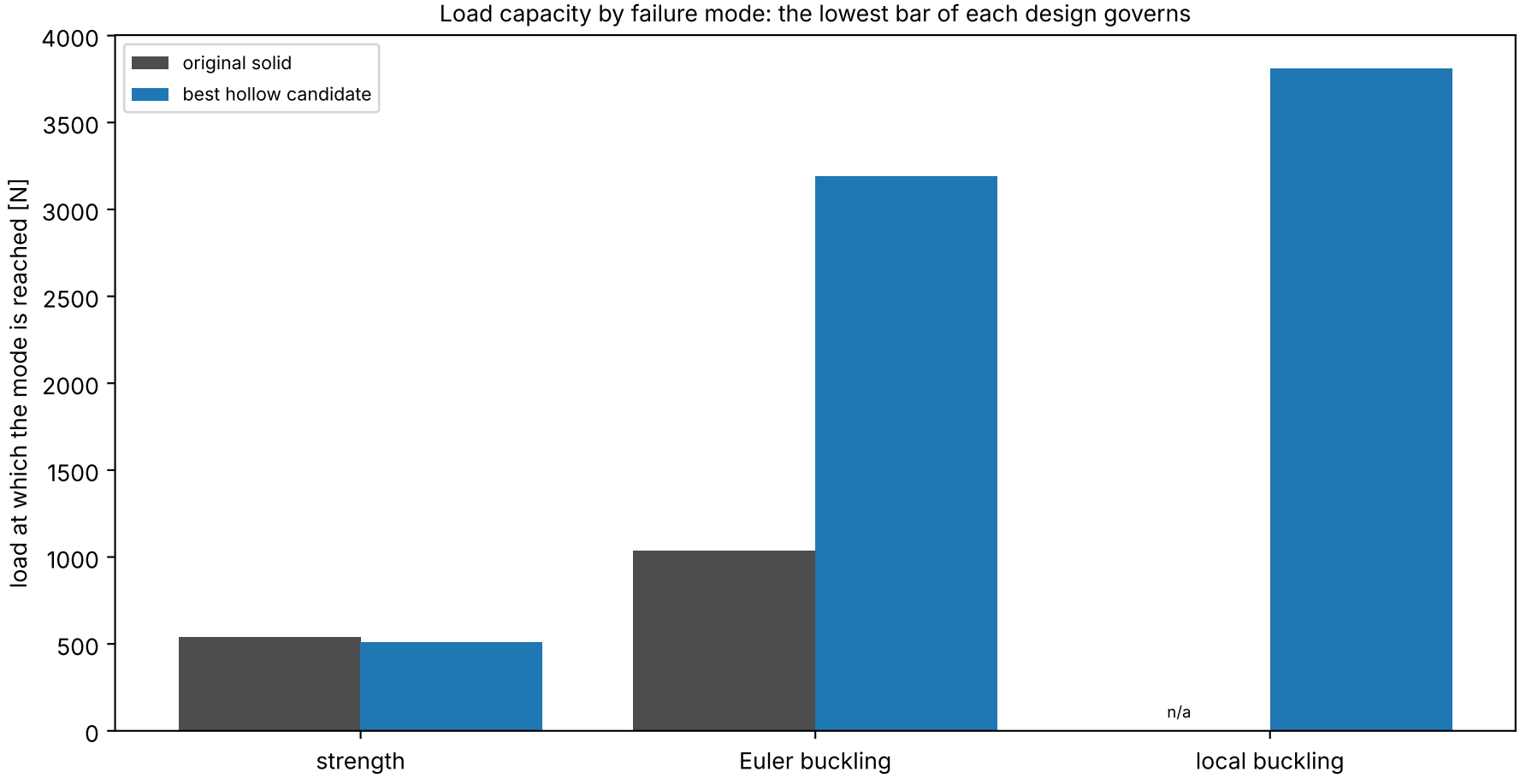

Shear-to-bending energy ratio per region (red = joint)
R = U_shear / U_bend (original solid)
share of strain energy [%] (dots)
region id, in rank order (made solid first, left)
6 5 14 13 1 4 11 12 2 15 3 10 8 9 7
Filled = hollow in the final design; marker area = strain-energy share
R
slenderness lambda = L / r_g (joints plotted at 2)

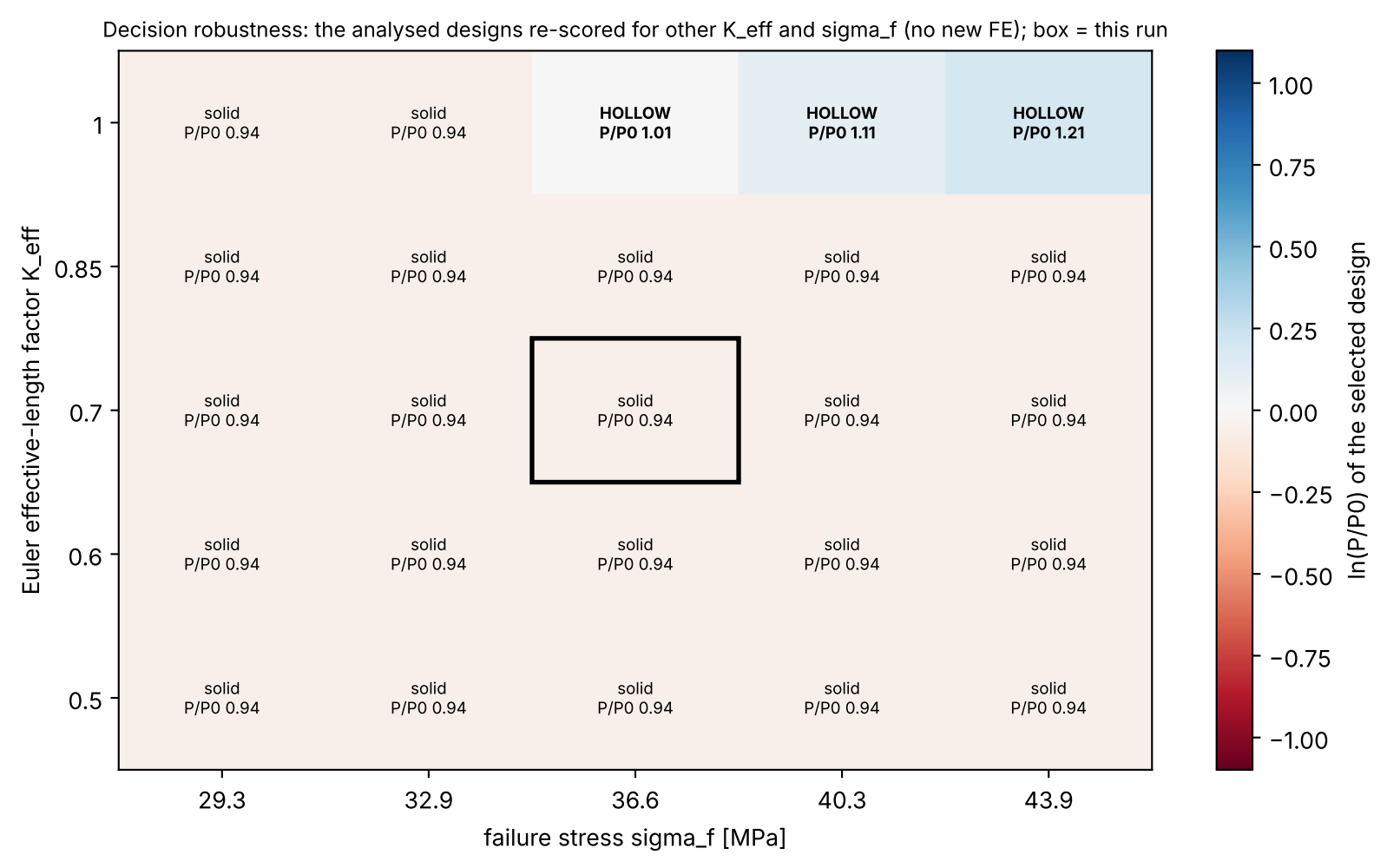

Constant-mass wall thickness for each design in the ranked family
outer offset = hollow wall s
as built (s - 0.2 mm)
t_min = 1.5 mm
analysed by FE
wall thickness [mm]
hollowed share of the part volume [%]
k = number of candidate regions kept solid (in rank order)
1.80
1.75
1.70
1.65
1.60
1.55
1.50
50
40
30
20
10
0
0
2
4
6
8
10
12
14

Regions (blue rows are hollow in the final design)

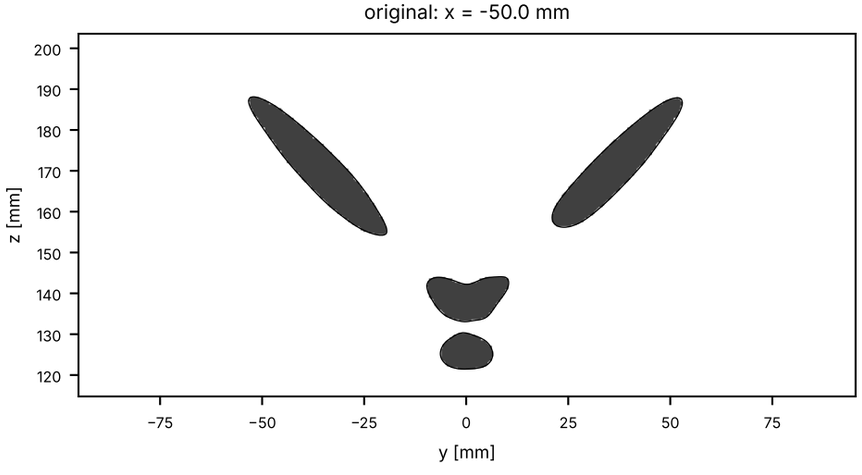
id	kind	rank	volume mm3	r mm	L mm	lambda	R	U share	final
1	joint	5	2.63e+03	3.86	-	-	0.528	4.8%	solid
2	joint	9	999	3.61	-	-	0.169	2.4%	solid
3	joint	11	1.01e+03	3.61	-	-	0.145	2.5%	solid
4	joint	6	867	2.83	-	-	0.489	1.8%	solid
5	joint	2	677	2.24	-	-	0.602	2.0%	solid
6	joint	1	326	2	-	-	0.79	0.8%	solid
7	strut	15	3.07e+03	4	63.1	31.6	0.0467	7.5%	solid
8	strut	13	3.58e+03	4	73.1	36.6	0.0676	10.8%	solid
9	strut	14	3.27e+03	4	72.5	36.3	0.0512	11.8%	solid
10	strut	12	321	3	8.83	5.89	0.0844	0.6%	solid
11	strut	7	507	2.83	13.6	9.61	0.176	1.4%	solid
12	strut	8	500	2.83	15.6	11.1	0.17	1.3%	solid
13	strut	4	1.04e+03	2.83	23	16.3	0.535	3.5%	solid
14	strut	3	1.03e+03	2.24	26	23.3	0.577	3.5%	solid
15	strut	10	167	2	7.24	7.24	0.168	0.4%	solid
16	mount	-	6.54e+03	5	-	-	0.497	7.5%	solid
17	mount	-	3.49e+03	4.12	-	-	0.964	17.8%	solid
18	mount	-	3.5e+03	4.12	-	-	1.04	17.7%	solid
19	mount	-	8.42e+03	5	-	-	1.07	1.8%	solid

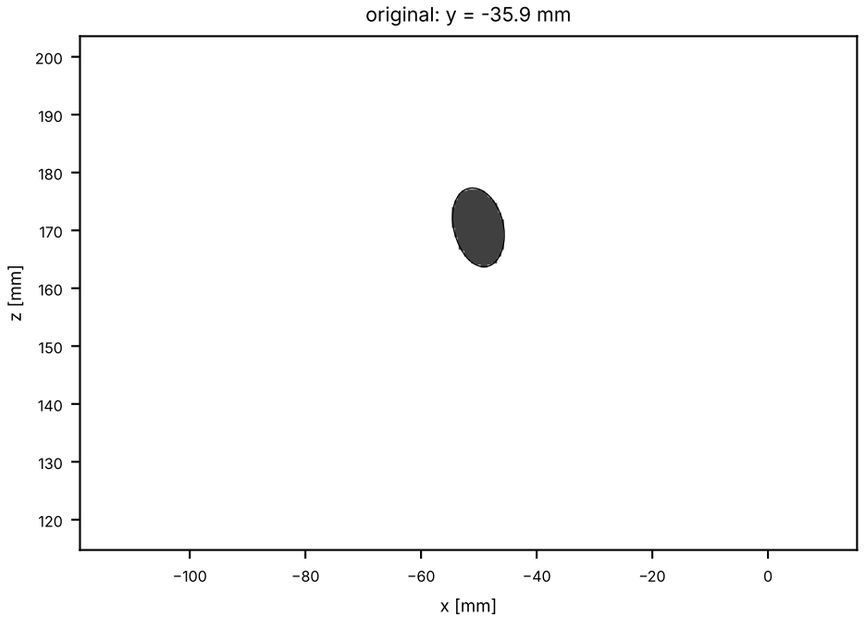

Performance of the analysed designs (governing mode: S strength, E Euler, L local)
K / K0 (stiffness)
P / P0 (load capacity)
stiffness floor
capacity floor
ratio to the original solid
1.05
1.00
0.95
0.90
0.85
S
0
2
4
6
8
10
12
14
k (regions kept solid)
Objective (blue = acceptable); best k = 1, rejected → solid kept
acceptance: ln(1 + 0.02)
J = [w_K ln(K/K0) + w_P ln(P/P0)] / (w_K + w_P)
0.02
0.01
0.00
−0.01
−0.02
−0.03
−0.04
k

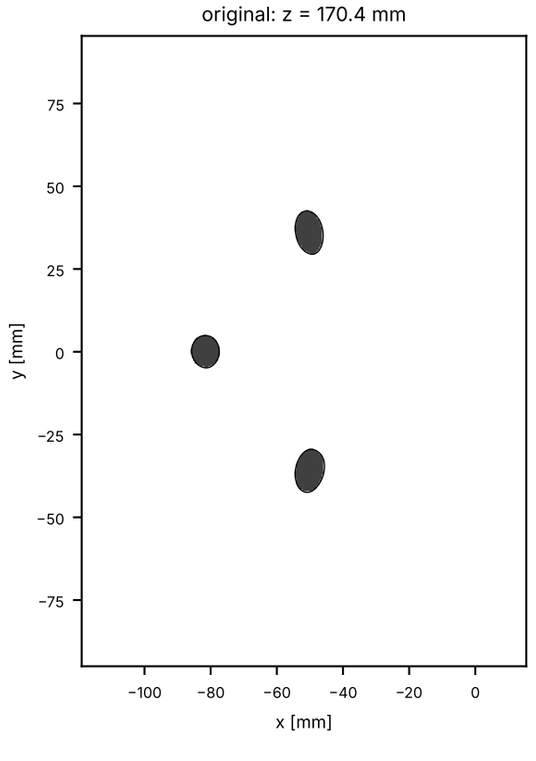

Load capacity by failure mode: the lowest bar of each design governs
original solid
best hollow candidate
load at which the mode is reached [N]
4000
3500
3000
2500
2000
1500
1000
500
0
n/a
strength
Euler buckling
local buckling

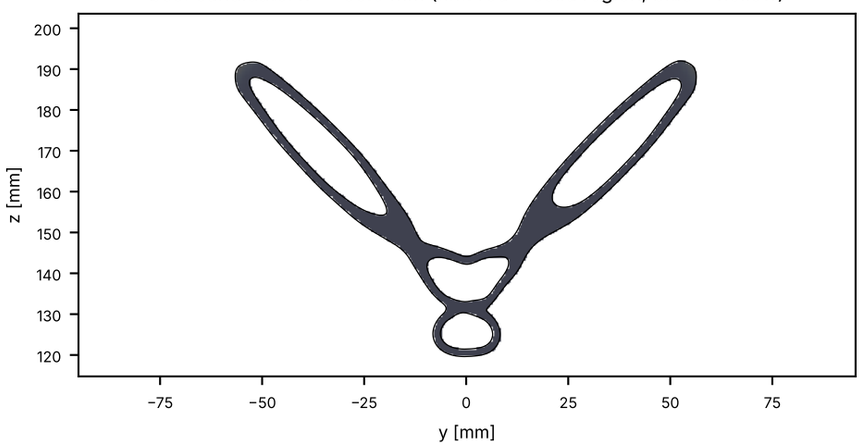

Decision robustness: the analysed designs re-scored for other K_eff and sigma_f (no new FE); box = this run
Euler effective-length factor K_eff
failure stress sigma_f [MPa]
ln(P/P0) of the selected design
1
0.85
0.7
0.6
0.5
29.3
32.9
36.6
40.3
43.9
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
solid P/P0 0.94
HOLLOW P/P0 1.01
HOLLOW P/P0 1.11
HOLLOW P/P0 1.21

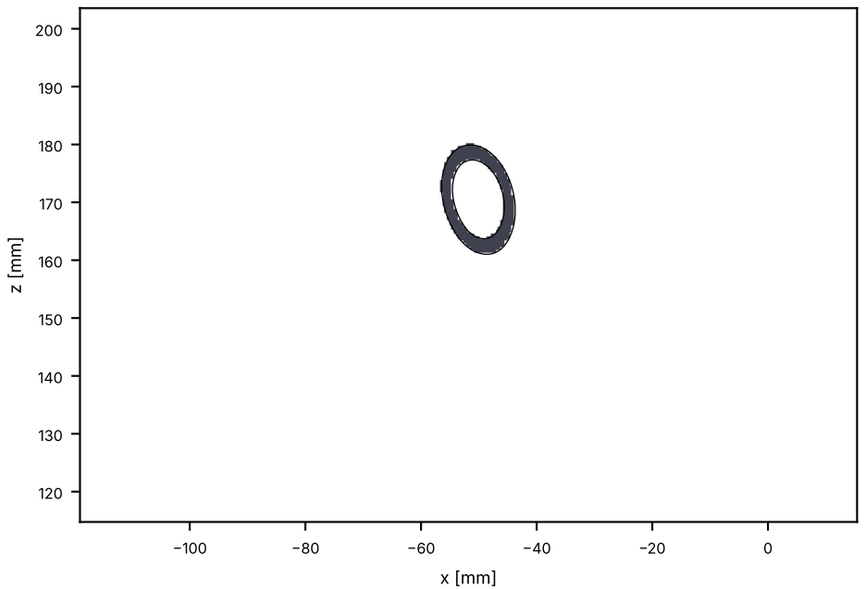
Sections through the largest hollow region of the best candidate (or the largest joint); material grey, void white

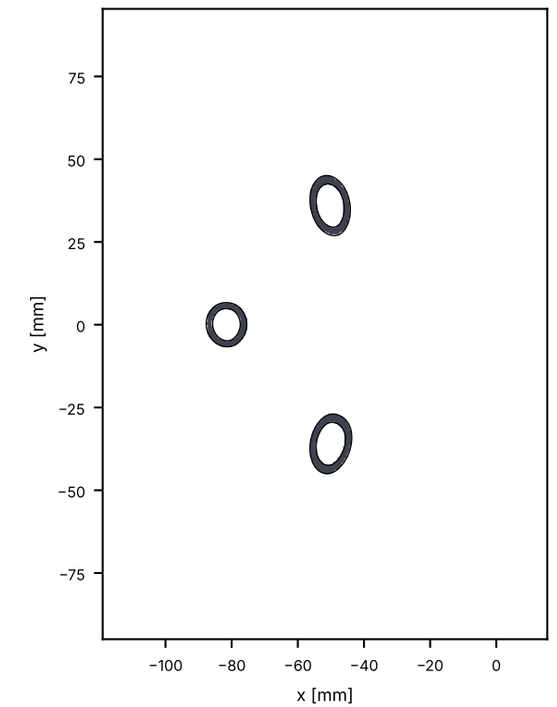




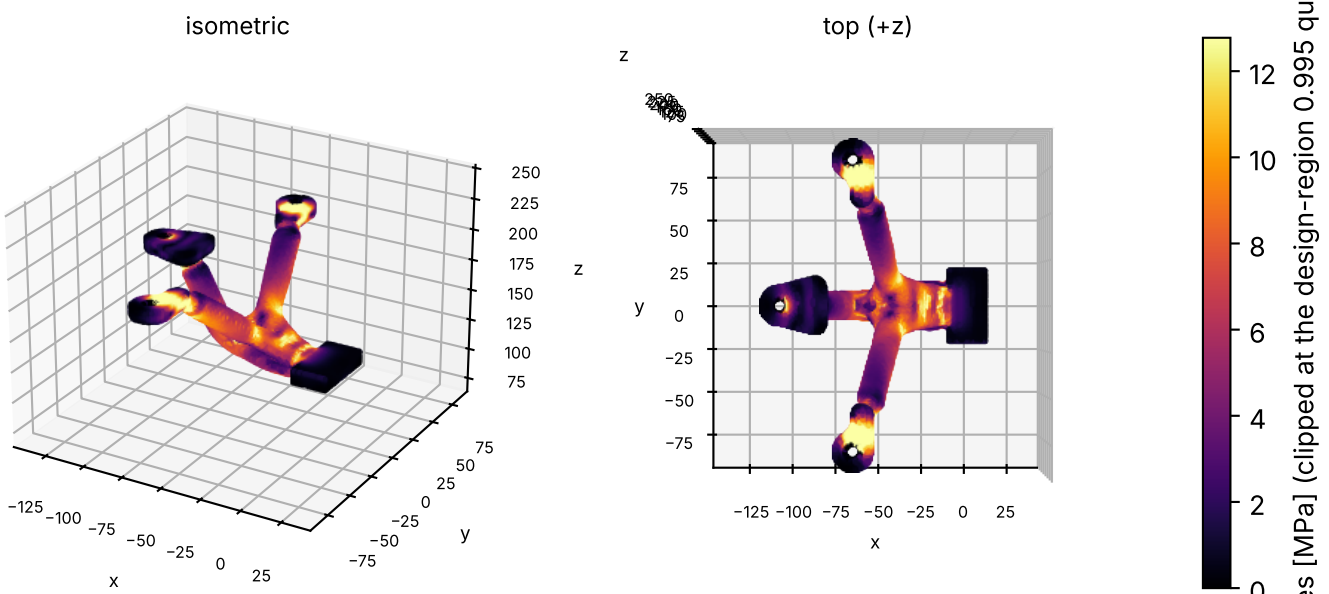


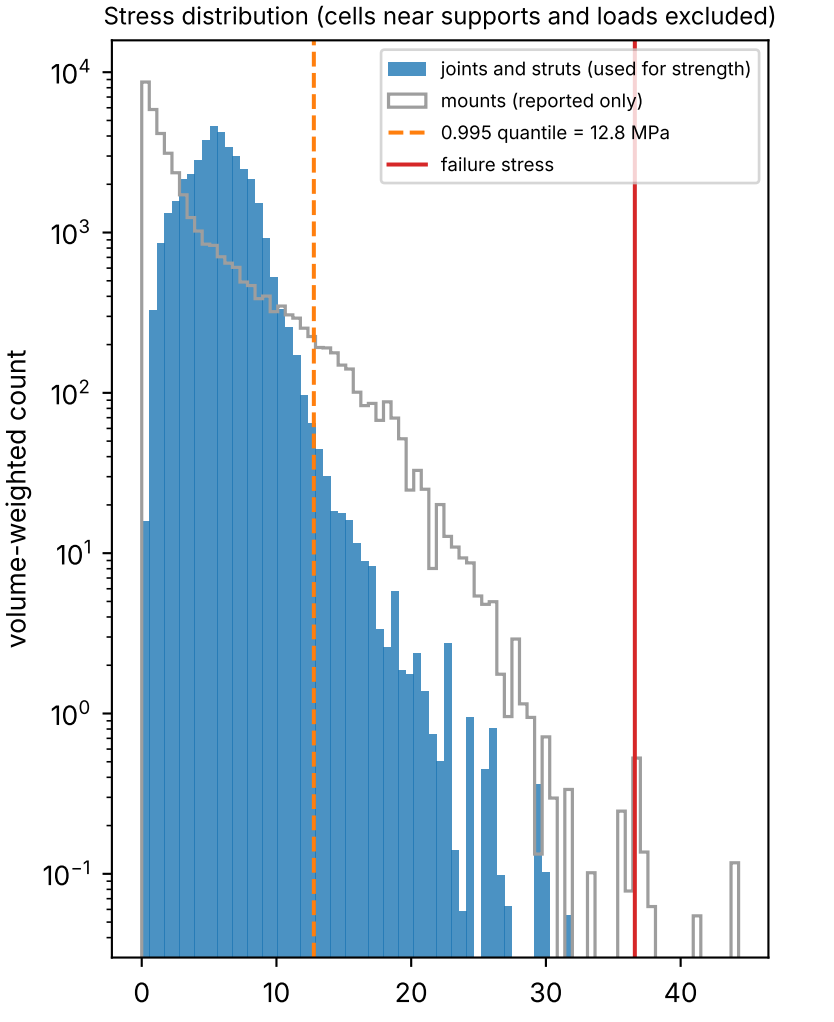






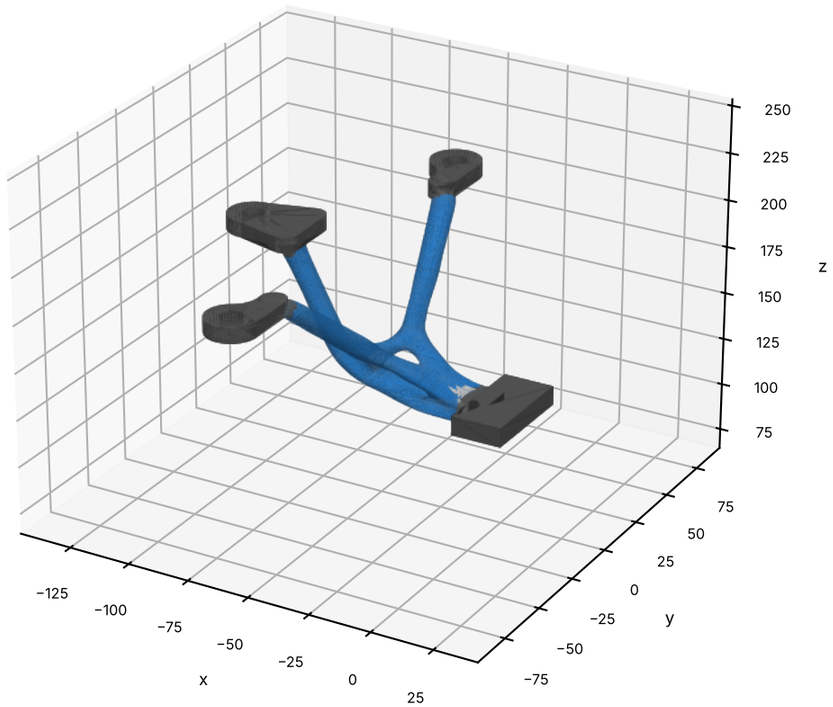
Best hollow candidate: von Mises stress at |F| = 200 N



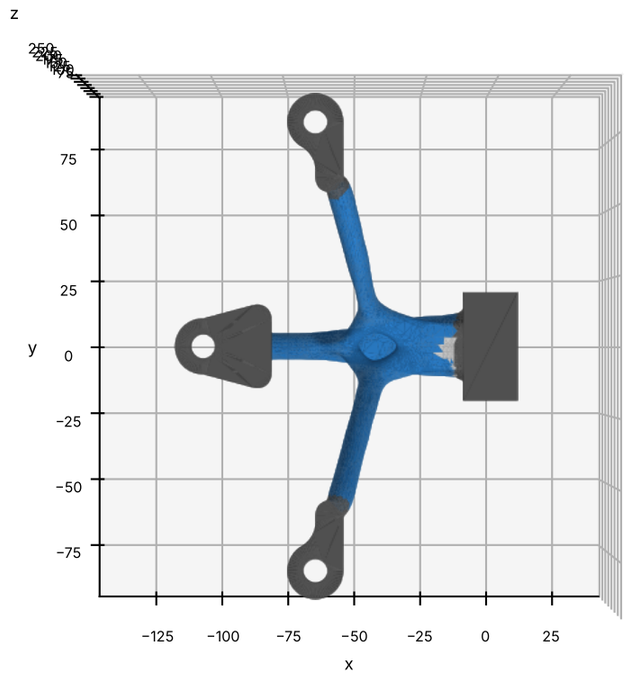


Best candidate: hollow regions blue (wall 1.81 mm), solid regions light grey, mounts dark grey - NOT adopted: final_design.stl is the original solid

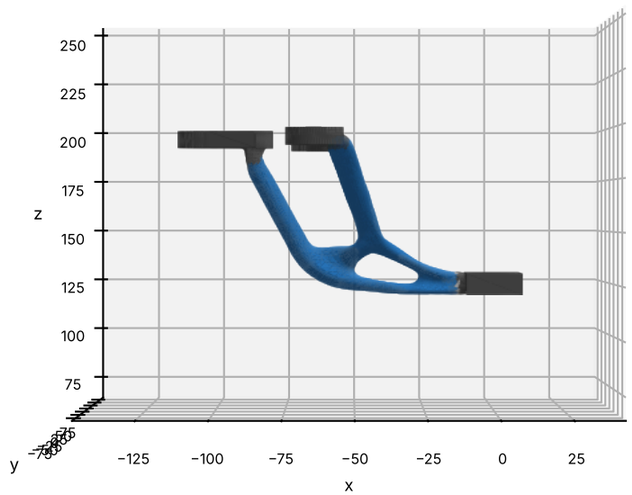
isometric



top (+z)



front (-y)

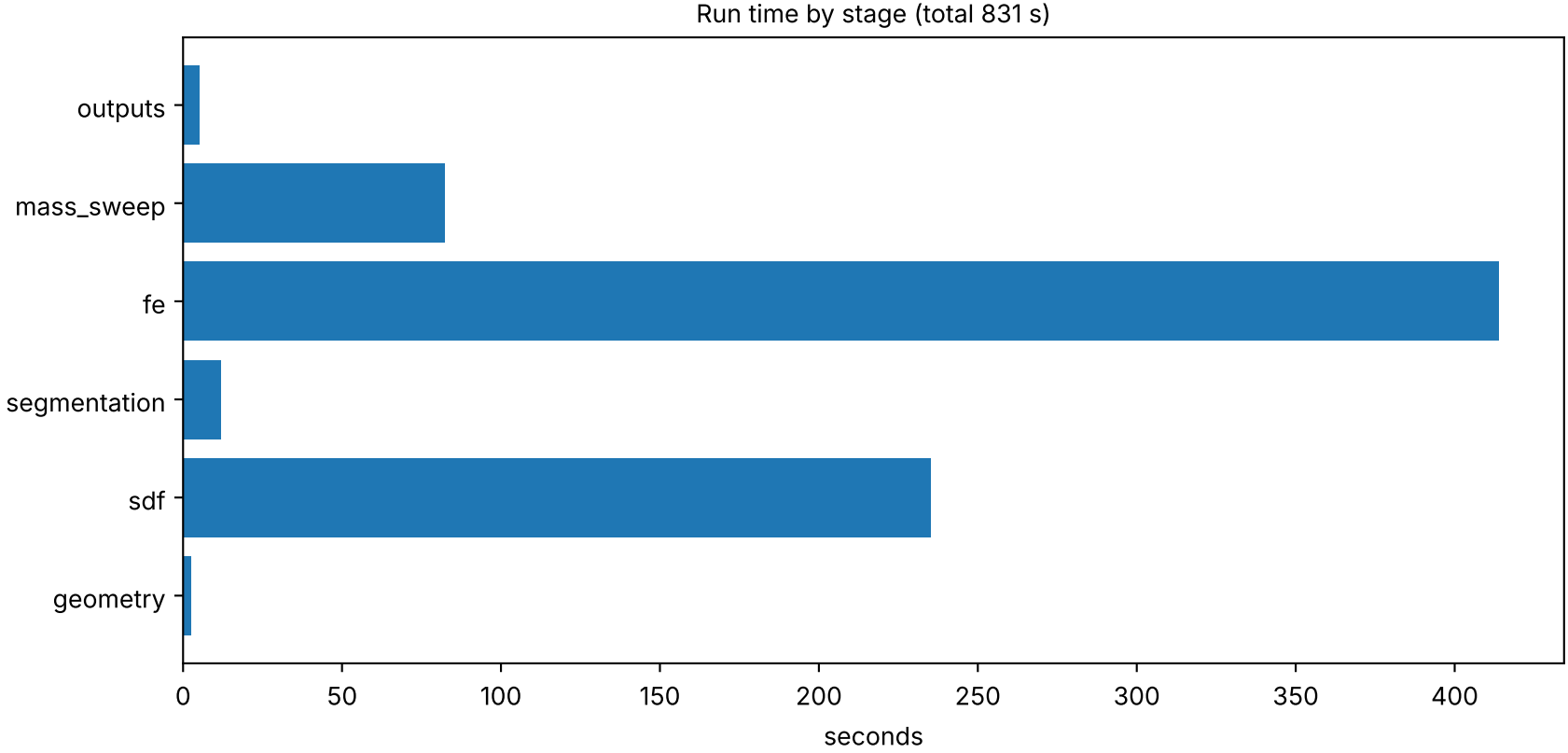


```
Enclosed voids of the best candidate: 1
  void 2: 18378 mm3

Small voids filled before analysis: 0.0 mm3

Suggested escape holes (lowest-stress wall points, outward direction n):
  void 2: (-26.4, 5.2, 129.4) n = (-0.77, -0.44, -0.47), von Mises 0.5 MPa, d = 3.0 mm
  void 2: (-45.2, -4.8, 140.2) n = (+0.64, +0.66, -0.39), von Mises 0.8 MPa, d = 3.0 mm

Escape holes are suggestions only: they are not cut into the mesh and their mass
(about pi d^2 t / 4 each) is not included in the constant-mass balance.
```


Run time by stage (total 831 s)
outputs
mass_sweep
fe
segmentation
sdf
geometry
0
50
100
150
200
250
300
350
400
seconds